



SPLICE PLAYGROUND

Visualize Splicing Into Structure

DNA editing, transcript consequences, and protein structure prediction in one continuous flow.

[Start](#)



2026S – Graduation Project2

Splice Playground

Make your own mutant

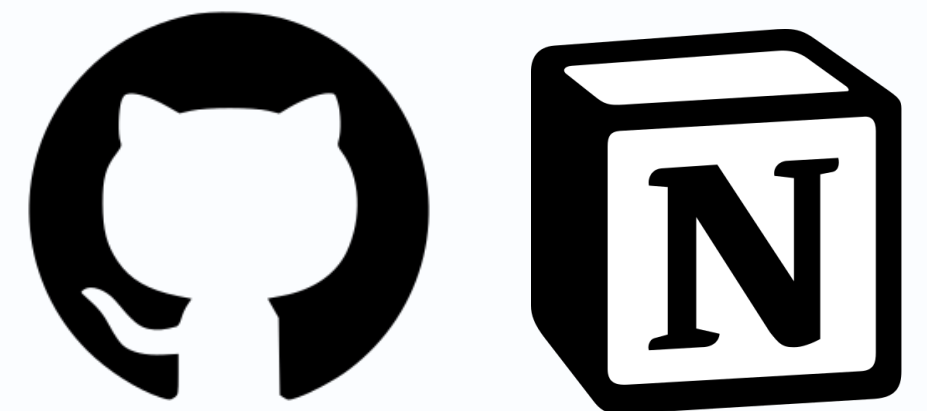
AI01분반

4조: 강민준, 김현우, 남윤정, 이정균, 최진범

Department of AI-Software

Outline

1. Concept Review
2. Splice AI training result
3. Service Architecture
4. DB Construction
5. Progress & Demo



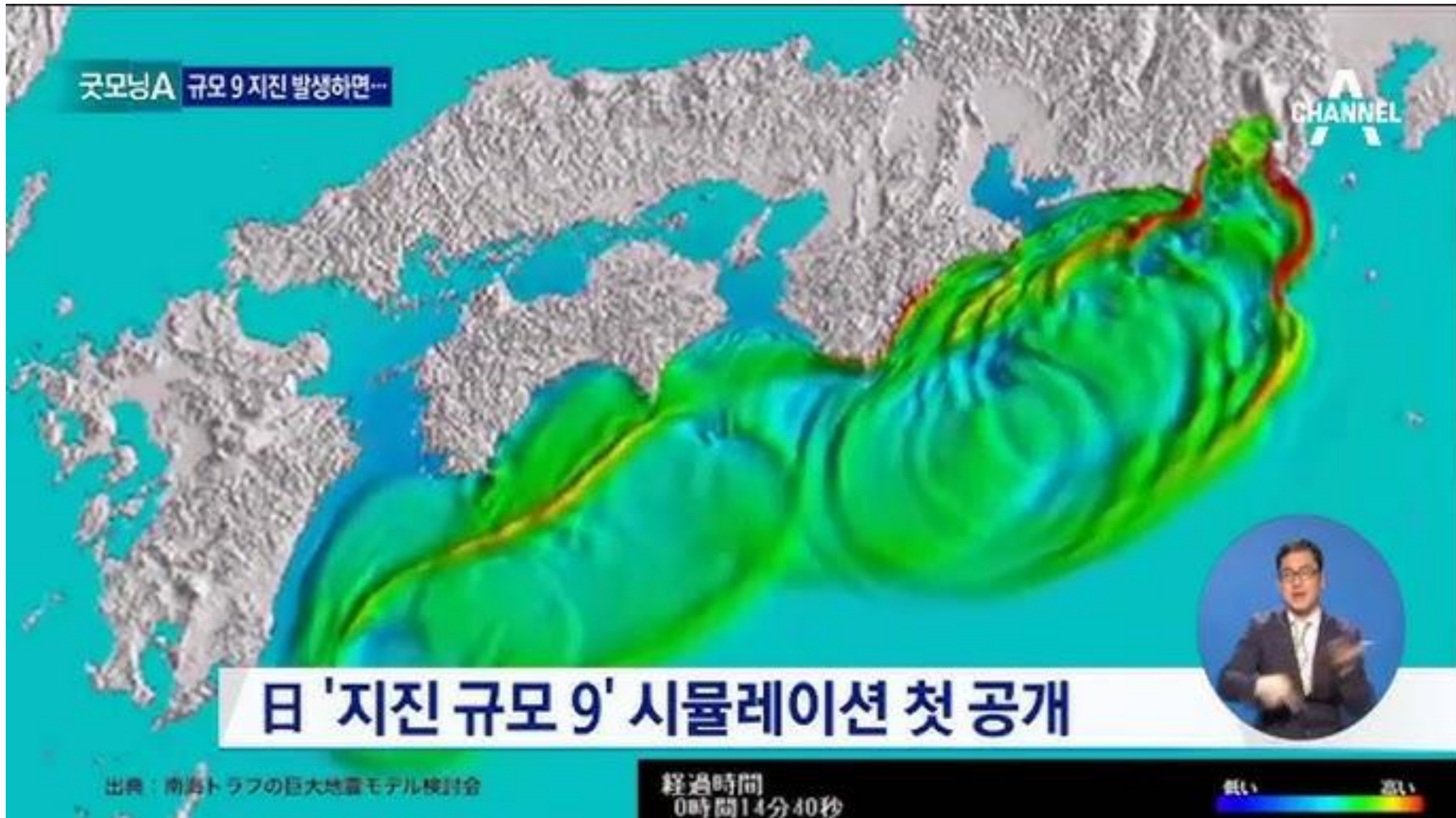


1. Concept Review

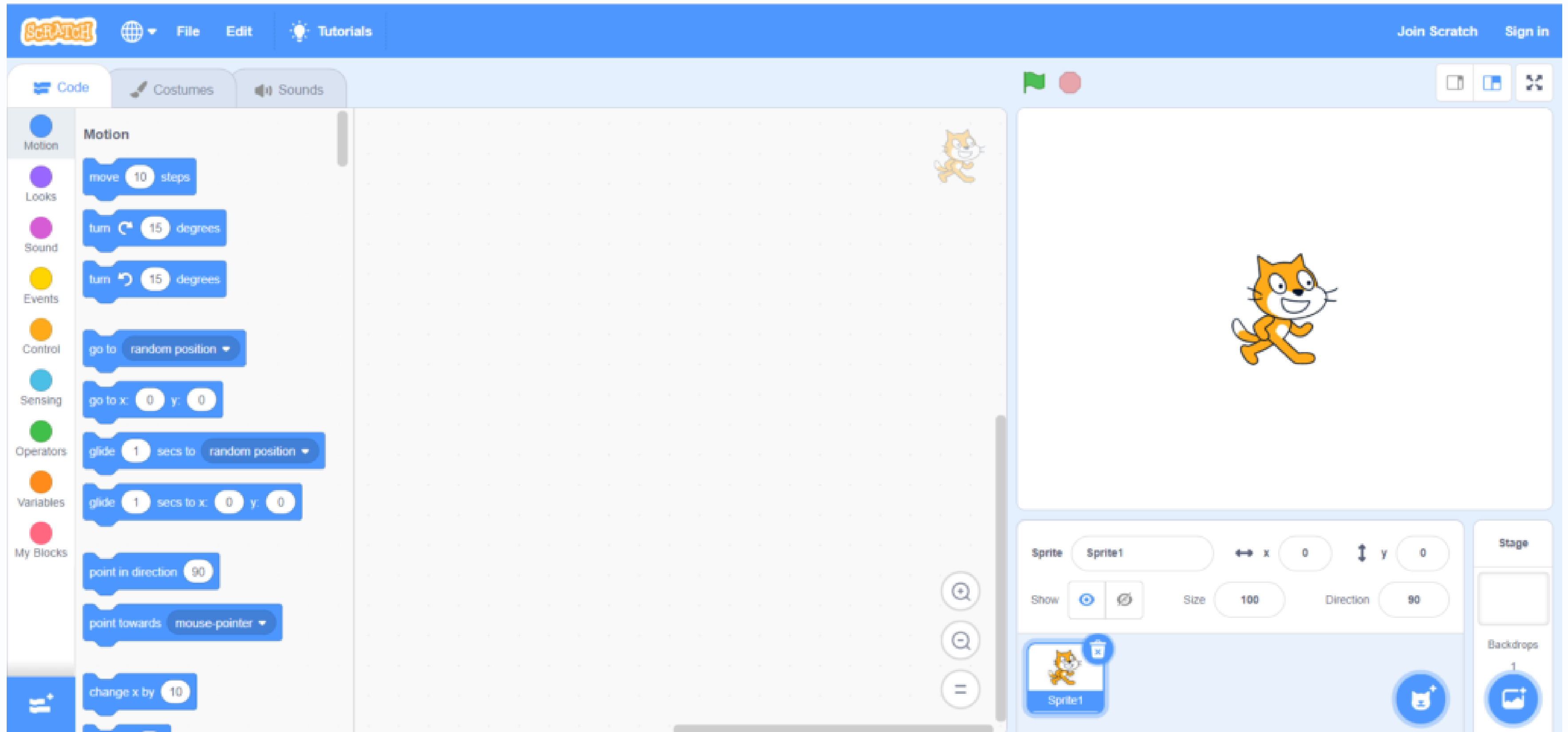
Concept Review



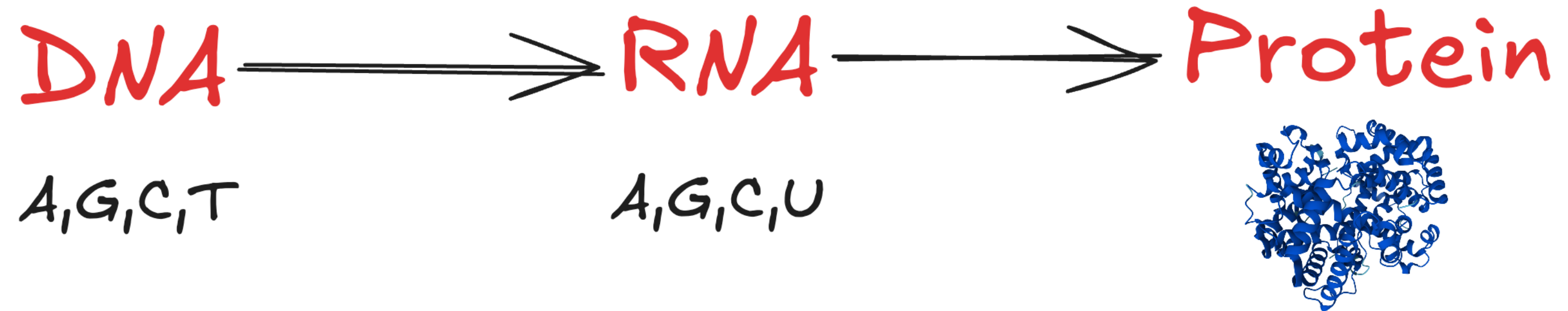
Concept Review



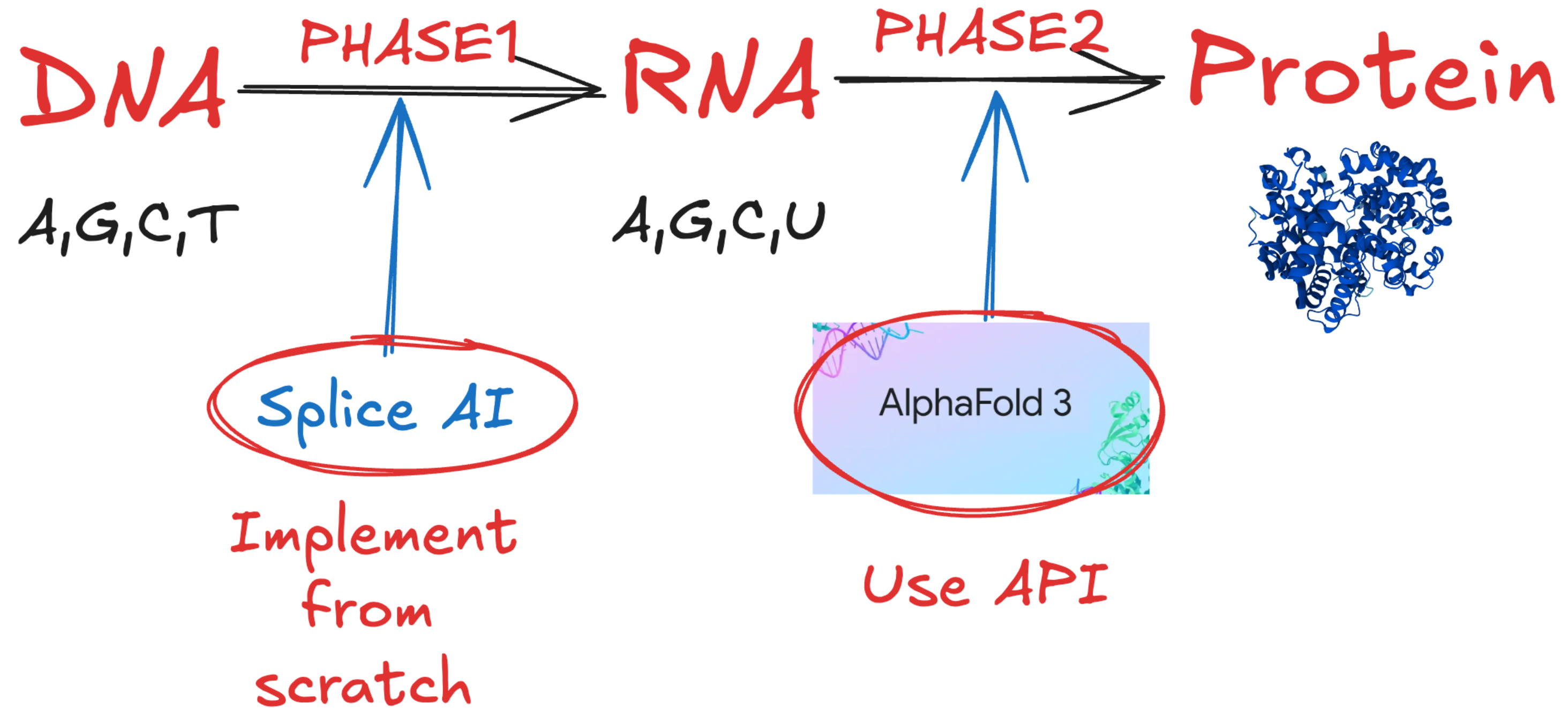
Concept Review



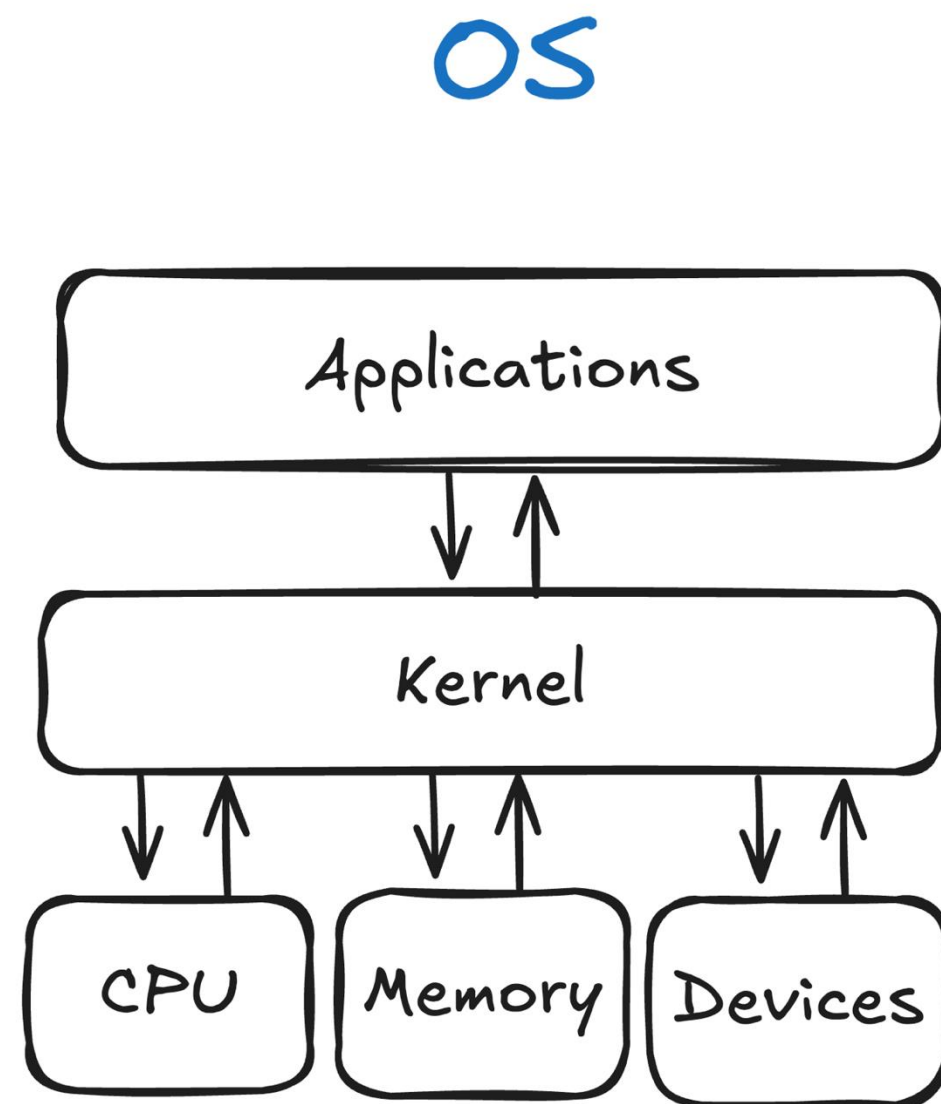
Concept Review



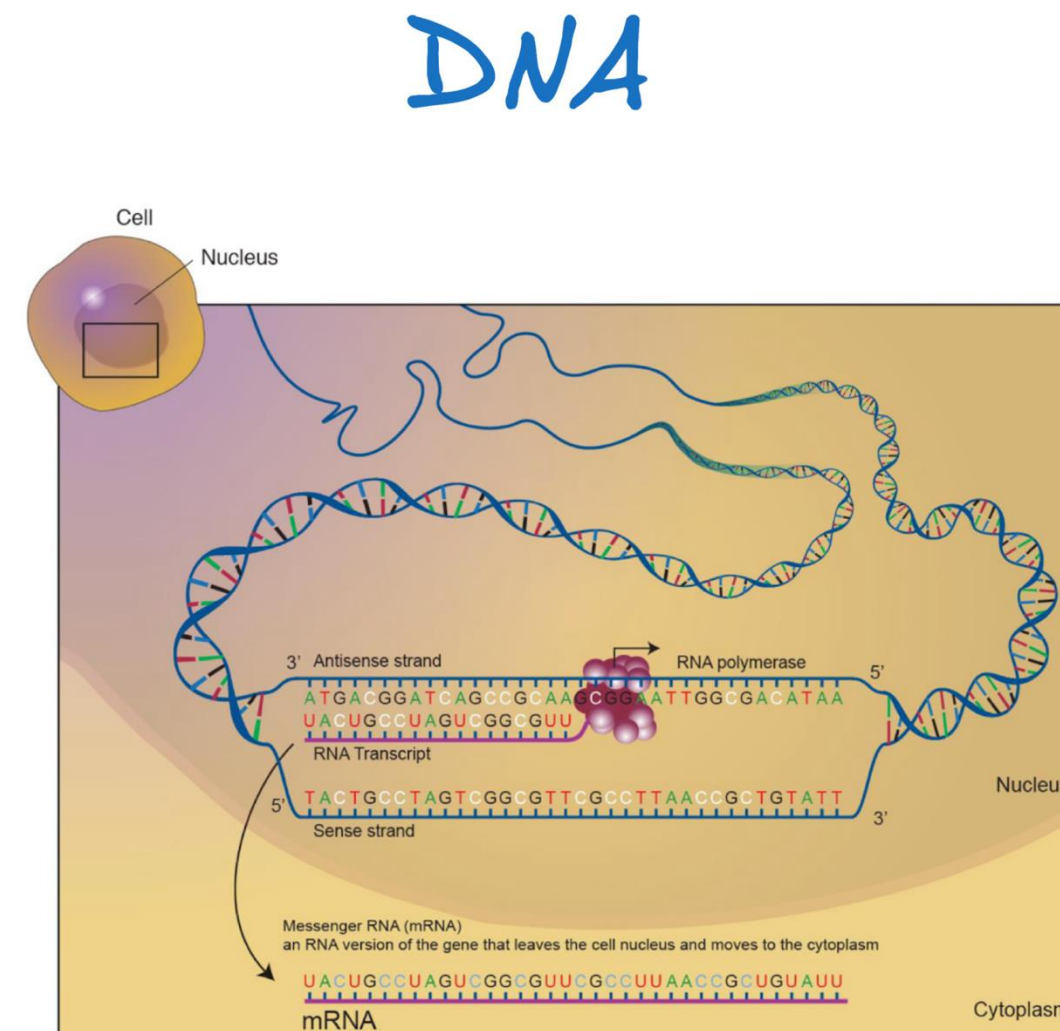
Concept Review



Why AI for Biological Sciences?



Systematic!!



Unsystematic...

What does mean “Unsystematic”?

DNA = kernel + interpreter + PYTHON + library +....

only with A, C, G, T

ATGACGGTCAGCCGCAACGGTACTTAAGATTTCGCC....

3.2billion characters!!...

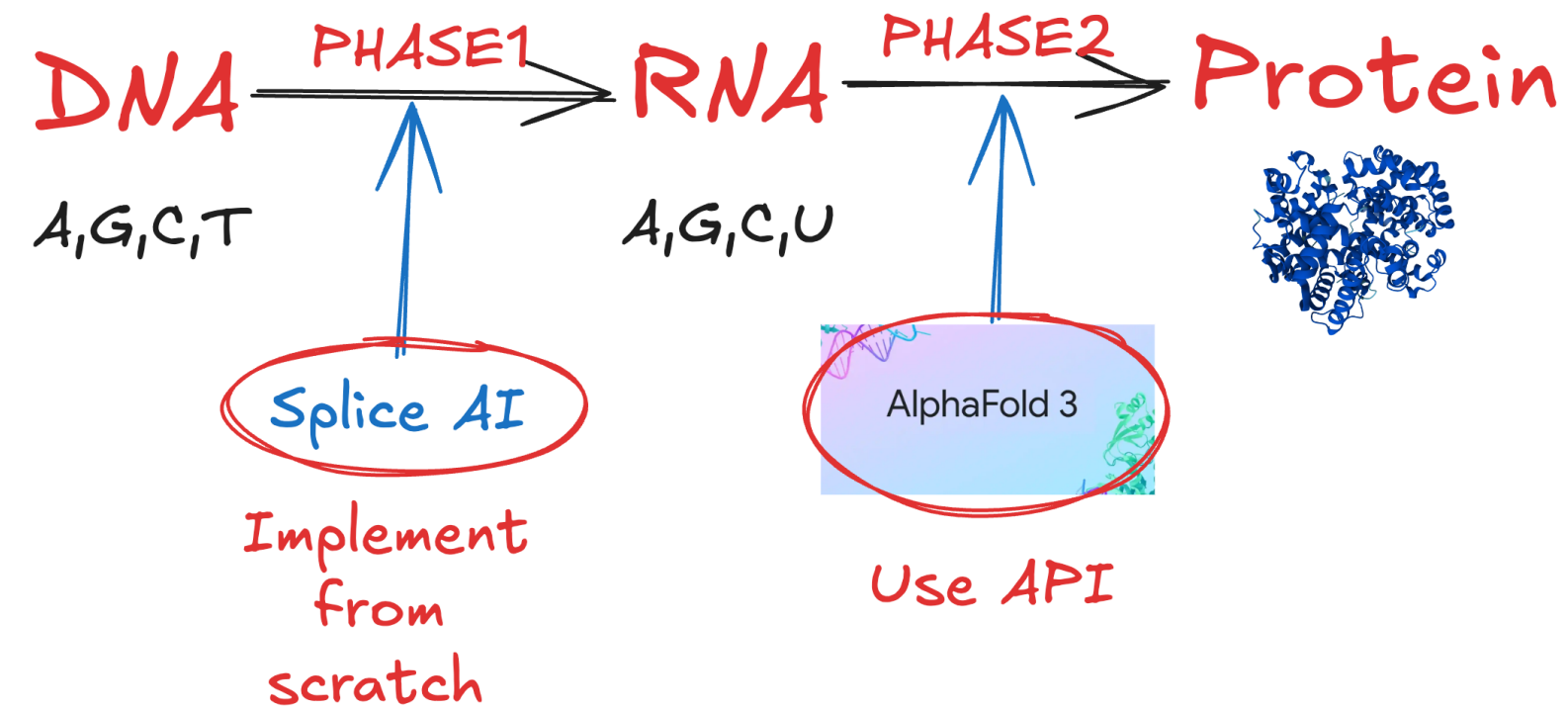
32억글자!!!...

Until today, human can't interpret DNA

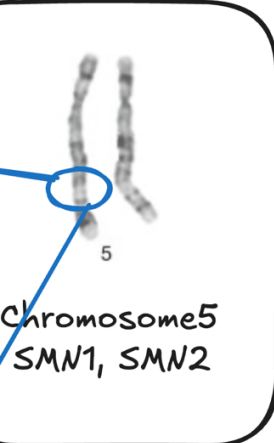
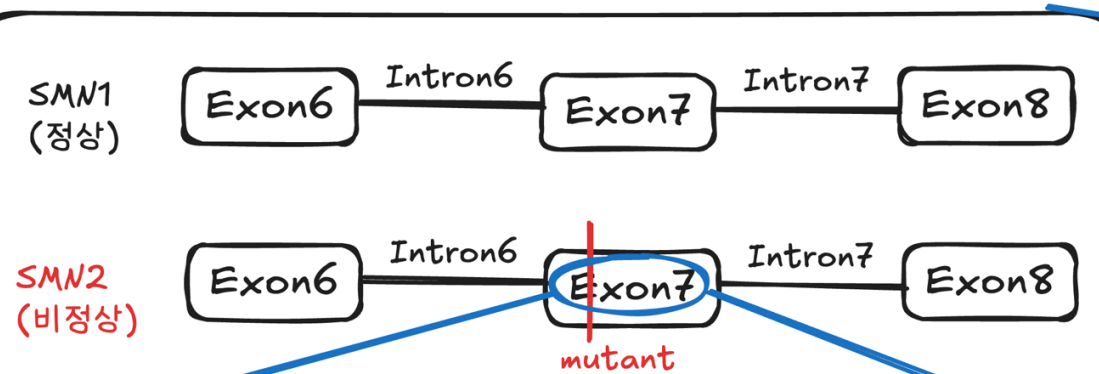


2. Splice AI Training Result

Splice AI



2. Manipulate DNA



User가
마음대로 조작 가능!

Next

Splice AI

3. Mature mRNA

SMN1(정상)



SMN2(비정상)



No Exon7 !

Make Protein

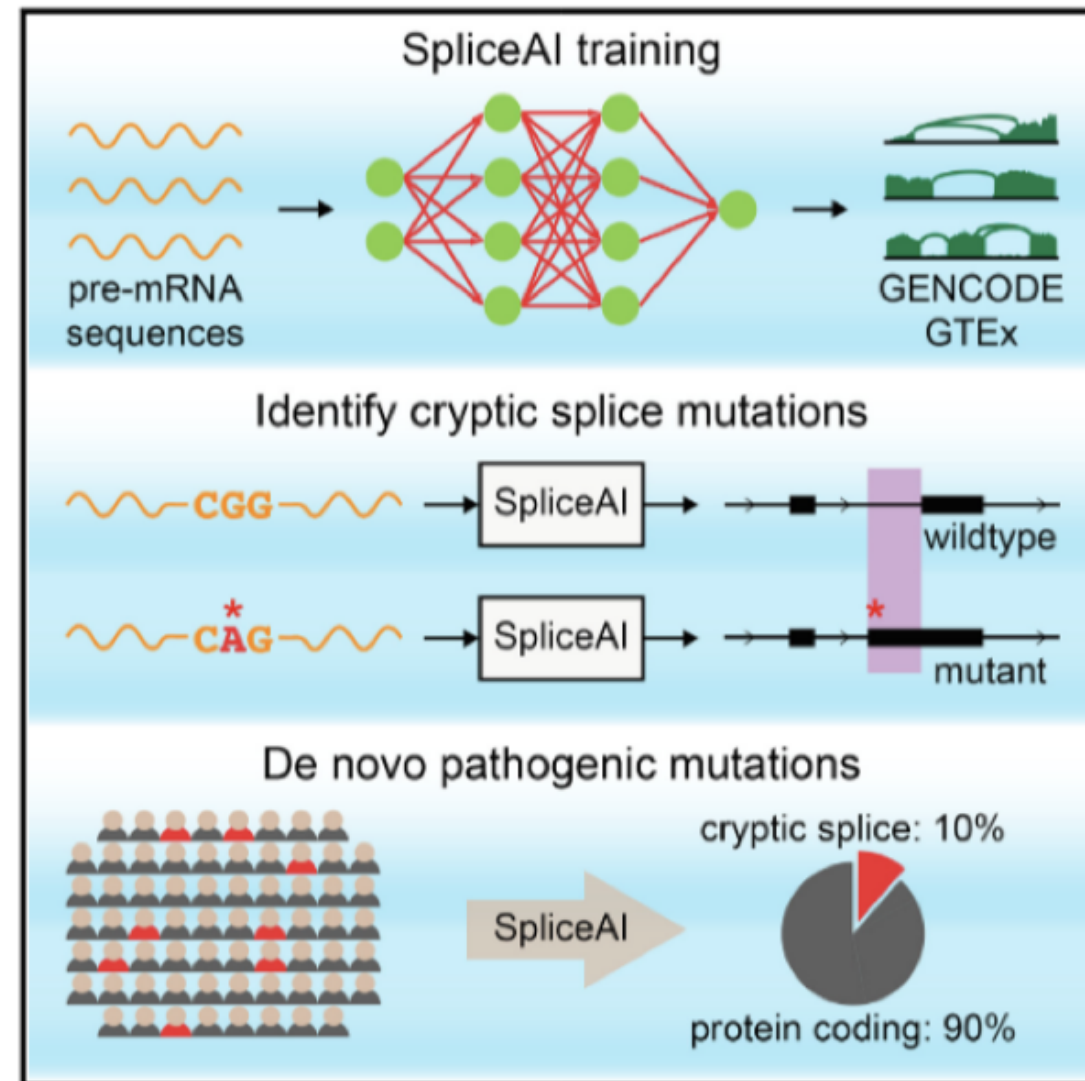
Splice AI

Article

Cell

Predicting Splicing from Primary Sequence with Deep Learning

Graphical Abstract



Authors

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Sofia Kyriazopoulou Panagiotopoulou,
Jeremy F. McRae, ..., Serafim Batzoglou,
Stephan J. Sanders, Kyle Kai-How Farh

Correspondence

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In Brief

A deep neural network precisely models mRNA splicing from a genomic sequence and accurately predicts noncoding cryptic splice mutations in patients with rare genetic diseases.

Highlights

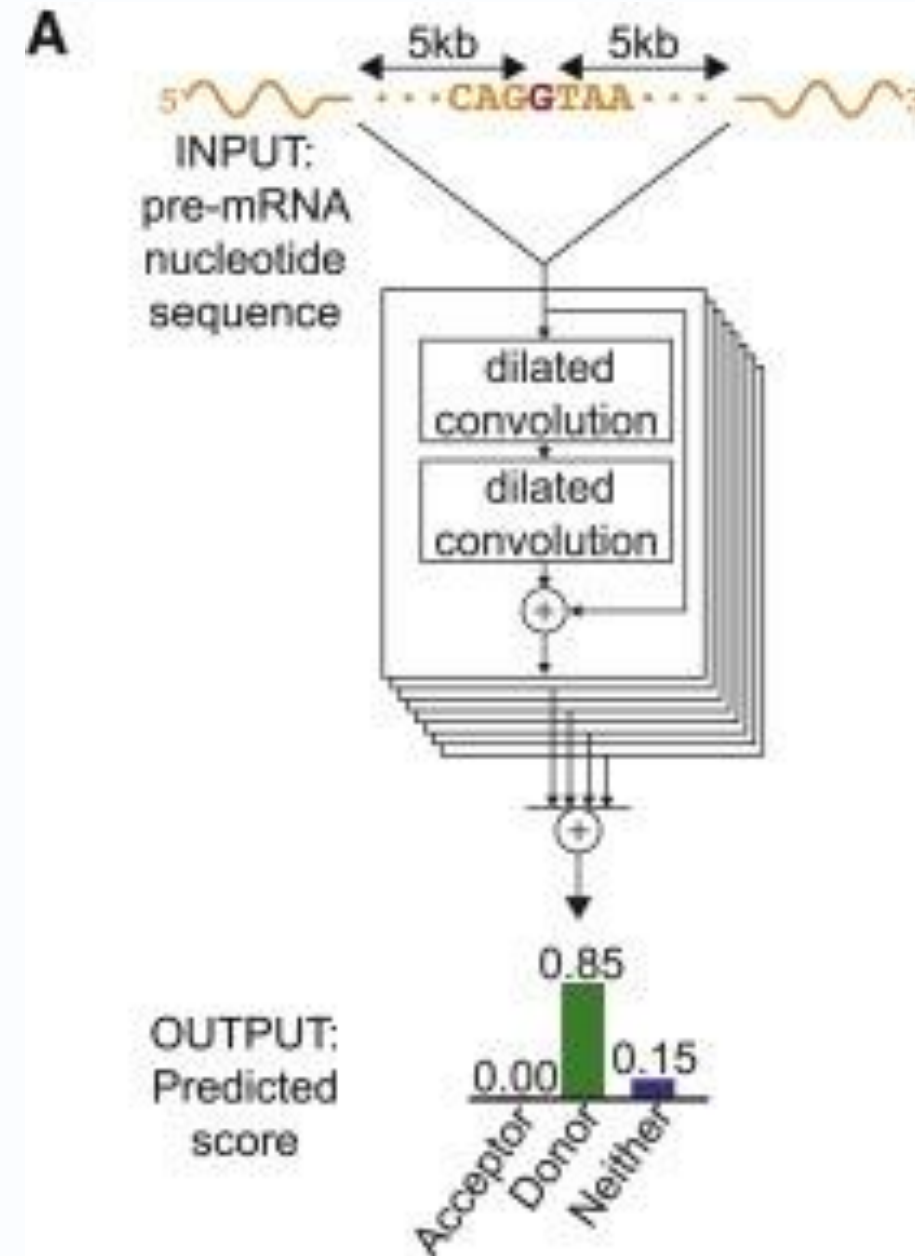
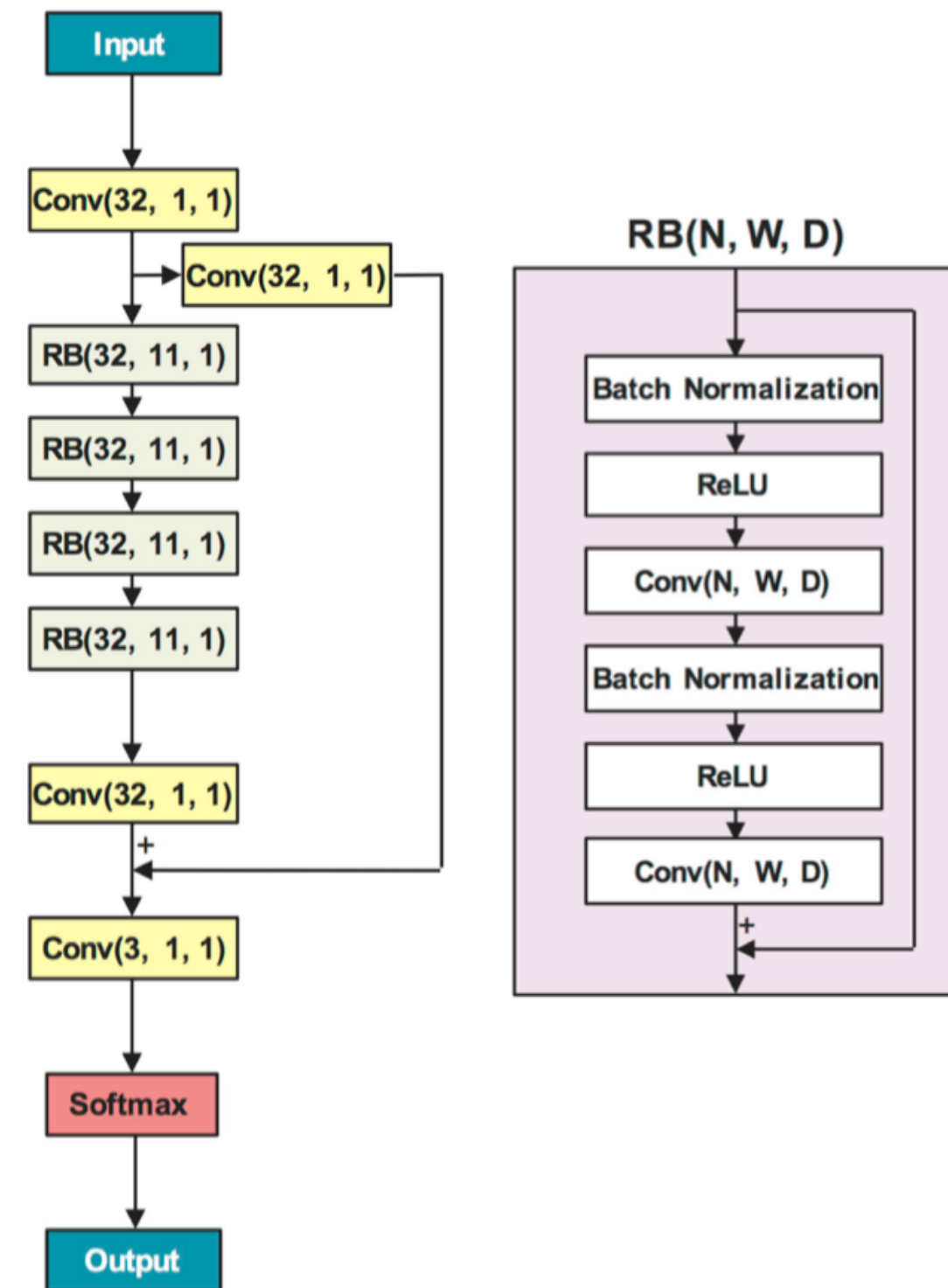
- SpliceAI, a 32-layer deep neural network, predicts splicing from a pre-mRNA sequence
- 75% of predicted cryptic splice variants validate on RNA-seq
- Cryptic splicing may yield ~10% of pathogenic variants in neurodevelopmental disorders
- Cryptic splice variants frequently give rise to alternative splicing



Jaganathan et al., 2019, Cell 176, 535–548
January 24, 2019 © 2018 Elsevier Inc.
<https://doi.org/10.1016/j.cell.2018.12.015>

Splice AI

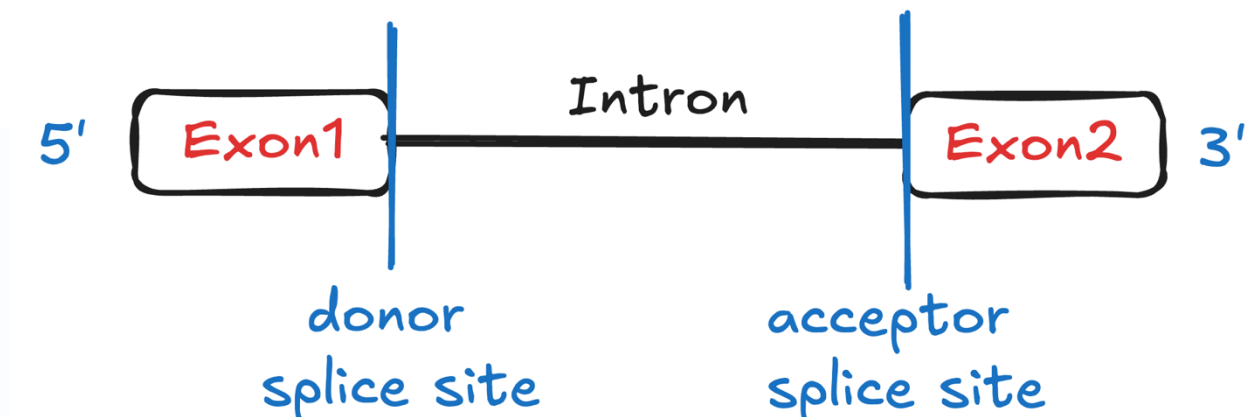
SpliceAI-80nt



- **Input:** one-hot encoded primary sequence

	A	T	G	G	C	T	C	A	T
A	[1,0,0,0]	1	0	0	0	0	0	1	0
C	[0,1,0,0]	0	0	0	1	0	1	0	0
G	[0,0,1,0]	0	0	1	1	0	0	0	0
T	[0,0,0,1]	0	1	0	0	1	0	0	1

- **Output:** donor / acceptor / neither (3 classes)



Studied SpliceAI paper at SNU Bio

교수

백대현 BAEK, DAEHYUN

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Email : baek@snu.ac.kr



Took 1st place at SNU

성 적 증 명 서

성 명 : 강민준
생년월일 : 2004. 6. 18.
학 번 : 2025- [REDACTED]

과 정 : 국내교환
수학대학 : 자연과학대학 생명과학부
수학기간 : 2025. 3. 3. ~ 2025. 6. 14., 2025. 9. 1. ~ 2025. 12. 12.

취 득 학 점

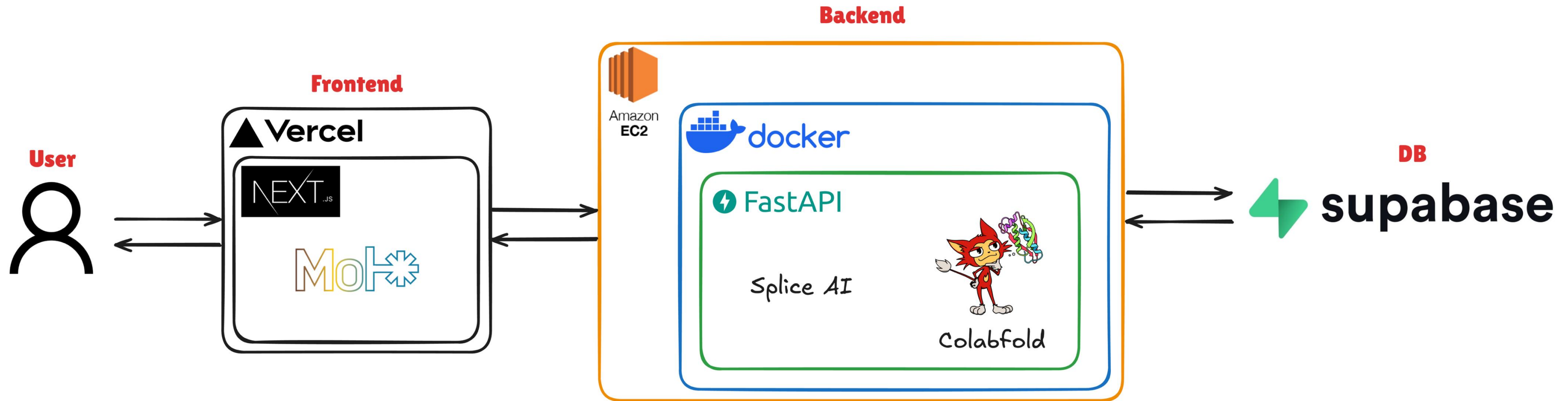
학 기	구분	교 과 목 명	학점	성적	학 기	구분	교 과 목 명
2025. 1.	교양	기초생물학 1	1	S			
2025. 1.	교양	생물학실험	1	A+			
2025. 1.	일선	인공지능 개론	3	S			
2025. 1.	대학원	영어논문작성법: 이공계열	1	S			
		학점계 : 6					
		평점평균 : 4.30					
2025. 2.	전선	생명과학을 위한 인공지능	4	A+			
		학점계 : 4					
		평점평균 : 4.30					
2025. 2.	전선	생명과학을 위한 인공지능	4	A+			
		평점평균 : 4.30					
		백점환산점수 : 100.00					

2025. 2. | 전선 | 생명과학을 위한 인공지능 | 4 | A+

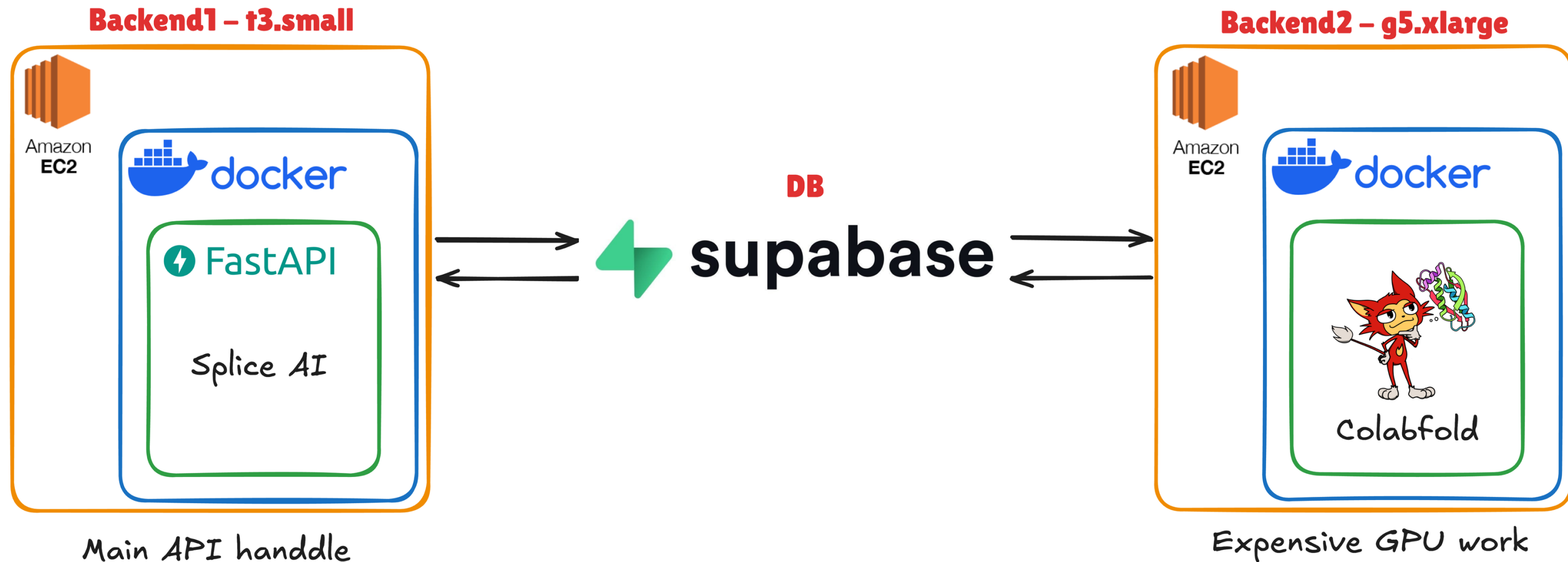


3. System Architecture

System Architecture



Backend details – Asynchronous Operation



t3.small

Family: **t3** 2 vCPU 2 GiB Memory Current generation: true
On-Demand RHEL base pricing: 0.0496 USD per Hour
On-Demand Windows base pricing: 0.0392 USD per Hour
On-Demand SUSE base pricing: 0.0518 USD per Hour
On-Demand Linux base pricing: 0.0208 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0243 USD per Hour

g5.xlarge

Family: **g5** 4 vCPU 16 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 1.013 USD per Hour
On-Demand Windows base pricing: 1.19 USD per Hour
On-Demand RHEL base pricing: 1.0636 USD per Hour
On-Demand SUSE base pricing: 1.0623 USD per Hour
On-Demand Linux base pricing: 1.006 USD per Hour

Backend details – Asynchronous Operation

Backend1 – t3.small



Amazon
EC2

- **CPU only**
- **API handle**



DB

supabase



Backend2 – g5.xlarge

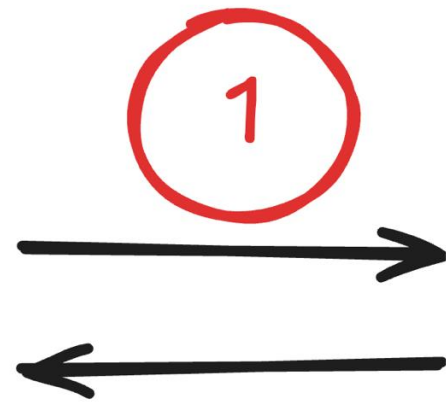


Amazon
EC2

- **on / off GPU worker!!**

Backend details – STEP1

Backend1 – t3.small

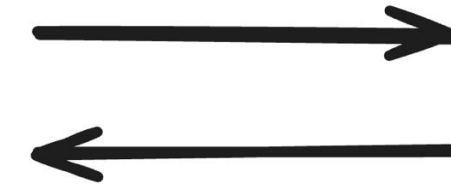


DB



supabase

Backend2 – g5.xlarge



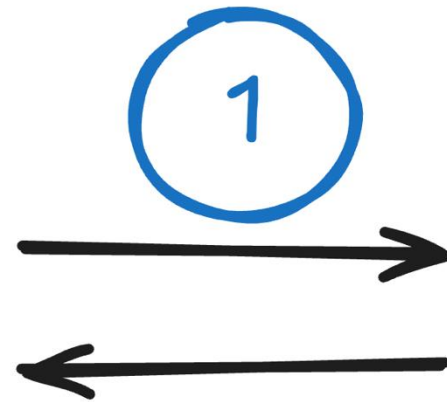
1. GPU연산이 필요 없는 STEP1~3에 대해 API handle (Splice AI 포함)
& STEP3에서 나온 결과인 '단백질 서열'을 DB에 업로드

Backend details – STEP2

Backend1 – t3.small



Amazon
EC2



DB



supabase

Backend2 – g5.xlarge

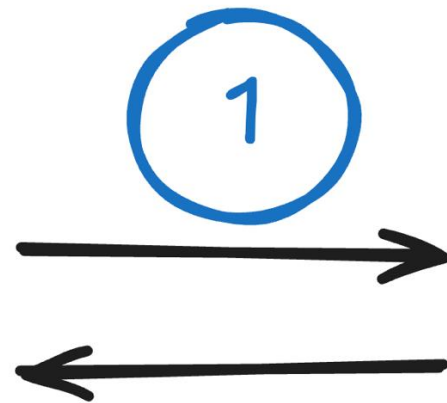


Amazon
EC2

2. DB에 올라온 '단백질 서열'을 읽어서 구조 예측(Colabfold!!)
& 예측한 3D 단백질 구조 파일 만들어서 다시 DB에 저장

Backend details – STEP3

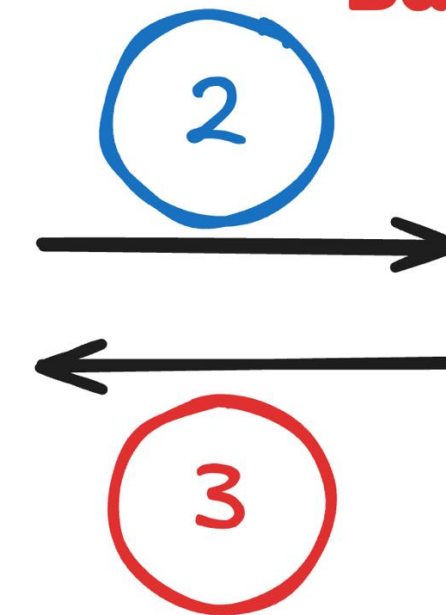
Backend1 – t3.small



DB

supabase

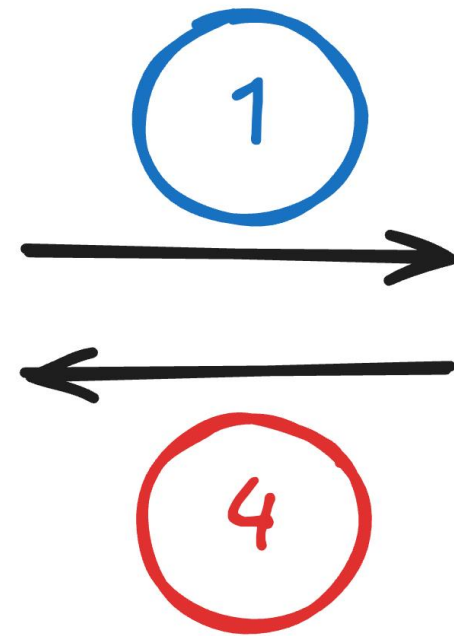
Backend2 – g5.xlarge



3. & 예측한 3D 단백질 구조 파일 만들어서 다시 DB에 저장

Backend details – STEP4

Backend1 – t3.small

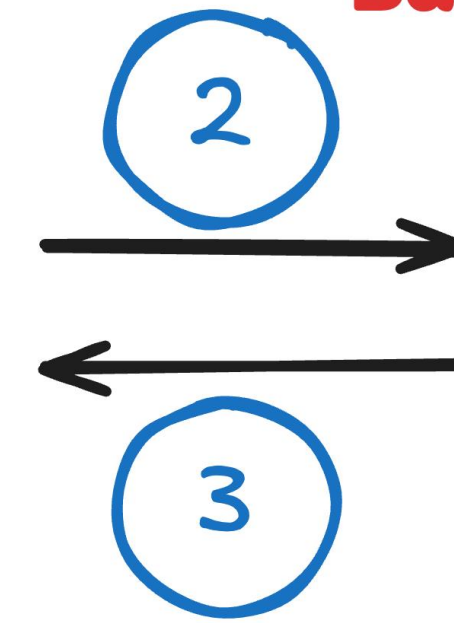


DB



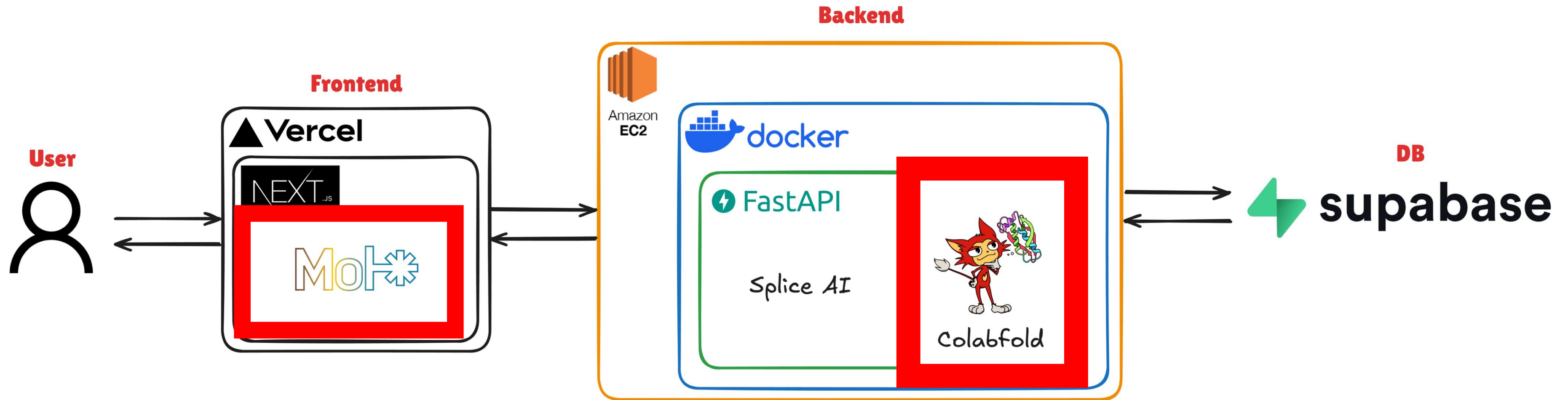
supabase

Backend2 – g5.xlarge



4. API handling하는 main 백엔드 서버가 단백질 구조 파일을 받아온 후,
프론트엔드에 단백질 구조 파일 반환

System Architecture



AlphaFold -> ColabFold

Brief Communication | [Open access](#) | Published: 30 May 2022

ColabFold: making protein folding accessible to all

[Milot Mirdita](#) ✉, [Konstantin Schütze](#), [Yoshitaka Moriwaki](#), [Lim Heo](#), [Sergey Ovchinnikov](#) ✉ & [Martin Steinegger](#) ✉

[Nature Methods](#) **19**, 679–682 (2022) | [Cite this article](#)

301k Accesses | **9013** Citations | **376** Altmetric | [Metrics](#)



SCIENCE
자 연 과 학 대 학 · 생 명 과 학 부

부교수

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Home

Download Structure

Source	PDB
PDB Id(s)	1tqn

Options

Apply

Download Density

Download File

Open Files

Load Trajectory

Download

Load Genome 3D (G3D)

Download Tunnels

Zenodo Import

Remote States

Zika+EM

P-450 Superposition & Validation

NPC

1RB8 Assembly Symmetry

Wed, 08 Apr 2026 02:28:52 GMT

Sequence of SARS-COV-2... Chain 1: Lipid bilayer 1

[1/1] 1/1/2022, 4:19:54 AM

13:57:39 Created Spacefill in 2ms.

13:57:39 Created S-protein trimers (glycosylated) in 0ms.

13:57:39 Created Spacefill in 2ms.

Structure Tools

Structure

SARS-COV-2_VIRION

Type Model

Nothing Focused

Measurements

Add

Quick Styles

Apply Representation

Default Cartoon Spacefill Surface

Apply Style

Default Illustrative

Components SARS-COV-2_VIRION

Preset Add

Lipid bilayer Spacefill

M-protein dimers Spacefill

E-protein pentamers

S-protein trimers (glycos...

Export Models

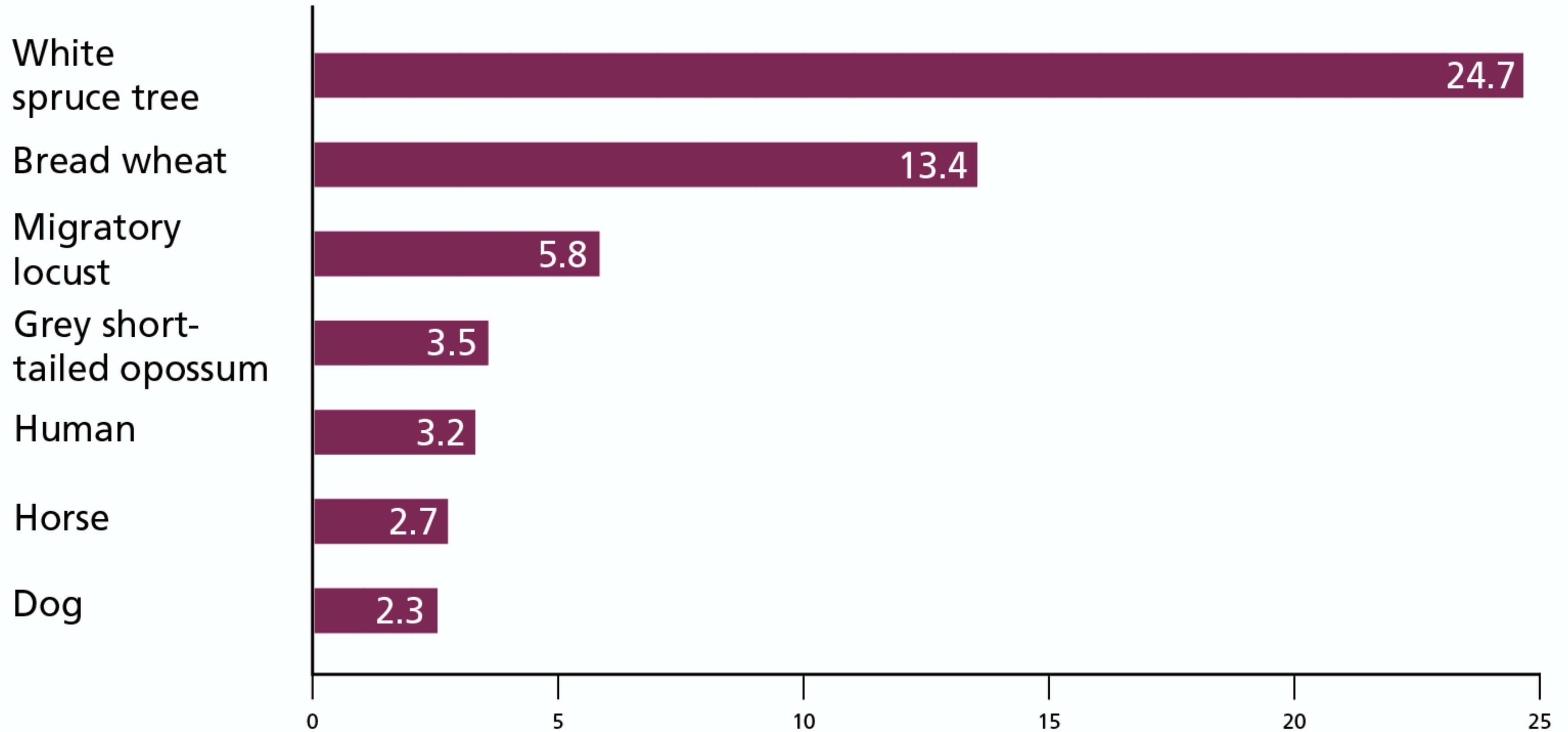
Export Animation

Export Geometry



4.DB Construction

Human Genome Size – 3.2billion!...



DB



DB upload pipeline

Raw data

FASTA file



3.2billion length(32억글자!..)

GTF file

gencode_v19_annotation_gtf[gencode_v19_annotation_gtf['gene_name']=='TERT']													
seqname	source	feature	start	end	score	strand	frame	gene_id	transcript_id	gene_type	gene_status	gene	
696990	chr5	HAVANA	gene	1253262	1295184	NaN	-	0	ENSG00000164362.14	ENST00000164362.14	protein_coding	KNOWN	
696991	chr5	HAVANA	transcript	1253262	1283125	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696992	chr5	HAVANA	exon	1282544	1283125	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696993	chr5	HAVANA	exon	1280273	1280453	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696994	chr5	HAVANA	exon	1279406	1279585	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696995	chr5	HAVANA	exon	1278756	1278911	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696996	chr5	HAVANA	exon	1266635	1268748	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696997	chr5	HAVANA	exon	1266579	1266650	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696998	chr5	HAVANA	exon	1260589	1260715	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
696999	chr5	HAVANA	exon	1258713	1258774	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	
697000	chr5	HAVANA	exon	1255402	1255526	NaN	-	0	ENSG00000164362.14	ENST00000484238.2	protein_coding	KNOWN	

table data

INPUT
specific gene that we want



UPLOADER

python™

DB



supabase

Supabase overview

Table Editor

schema public

+ New table

Search tables...

baseline_result

disease

editing_target_window

gene

protein_reference

protein_structure...

region

splice_altering_snv

structure_job

user_state

user_state_result

Filter by region_id, gene_id, region_type... or ask AI

Sort

Add RLS policy

Index Advisor

Enable Realtime

Role postgres

Insert

	region_id text	gene_id text	region_type text	region_number int4
	BRCA1_exon_1	BRCA1	exon	1
	BRCA1_exon_10	BRCA1	exon	10
	BRCA1_exon_11	BRCA1	exon	11
	BRCA1_exon_12	BRCA1	exon	12
	BRCA1_exon_13	BRCA1	exon	13
	BRCA1_exon_14	BRCA1	exon	14
	BRCA1_exon_15	BRCA1	exon	15
	BRCA1_exon_16	BRCA1	exon	16
	BRCA1_exon_17	BRCA1	exon	17
	BRCA1_exon_18	BRCA1	exon	18
	BRCA1_exon_19	BRCA1	exon	19
	BRCA1_exon_2	BRCA1	exon	2
	BRCA1_exon_20	BRCA1	exon	20
	BRCA1_exon_21	BRCA1	exon	21
	BRCA1_exon_22	BRCA1	exon	22

Page 1 of 4

100 rows

362 records

Data

Definition

GeneBe

- History
- Genes
- Annotate multiple
- Tools (Liftover, HGVS)
- Annotations Hub new
- About
- Contact

Genome Browser

Viewing: 17 : 43,063,833 - 43,063,913 (80 bp) [edit](#) Cursor: 43,063,841

<

>

-

+

☒ Only MANE transcripts



ClinVar

ClinVar

"TSC2"[GENE]

Search

Create alert
Advanced

Home
About
Access
Help
Submit
Statistics
FTP

We've updated the full submission template; previous versions are no longer supported. Download the latest version from [the ClinVar FTP site](#) and [read more about the available templates](#).

Classification type

clear

☐ Germline (6)
 ☒ Somatic (10)

Germline classification

clear

☐ Conflicting classifications (1)
 ☐ Benign (0)
 ☐ Likely benign (0)
 ☐ Uncertain significance (0)
 ☐ Likely pathogenic (2)
 ☐ Pathogenic (5)

Types of conflicts

clear

☐ P/LP vs LB/B (0)
 ☐ P/LP vs VUS (0)
 ☐ VUS vs LB/B (1)

Molecular consequence

☐ Frameshift (1)
 ☐ Missense (6)
 ☐ Nonsense (1)
 ☐ Splice site (0)
 ☐ ncRNA (4)
 ☐ Not provided (2)

Graphical view of search results

GRCh37

Gene

Pathogenic

Likely pathogenic

Uncertain significance

Likely benign

Benign

Conflicting

Not provided

other

2100K

2105K

2110K

2115K

2120K

2125K

2130K

2135K

Search results

Display options
Sort by Location
Download

Items: 10

Filters activated: Somatic. [Clear all](#) to show 12261 items.

Variation	Gene	Type	Condition	Classification	Review
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			
	TSC2	Pathogenic			

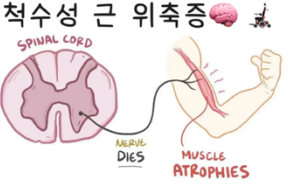
질병 선정에 도움을 주신 서울대 생명과학부 백대현 교수님,
박사과정 황현서 선배님, 포닥 전현성 선배님께 감사드립니다.



5. Progress

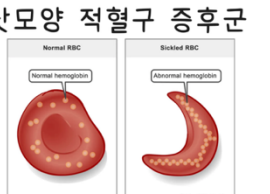
User Interface Review

1. Select Mutant



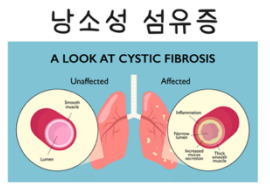
척수성 근 위축증
SPINAL CORD
MUSCLE ATROPHIES

Spinal Muscular Atrophy



낫모양 적혈구 증후군
Normal RBC
Sickled RBC

sickle cell anemia



낭소성 섬유증
A LOOK AT CYSTIC FIBROSIS
Unaffected
Affected

Cystic Fibrosis

뒤센 근이영양증

Image

Duchenne Muscular Dystrophy

가족성 고콜레스테롤혈증

Image

Familial Hypercholesterolemia

망막색소변성증

Image

Retinitis Pigmentosa

2. Manipulate DNA

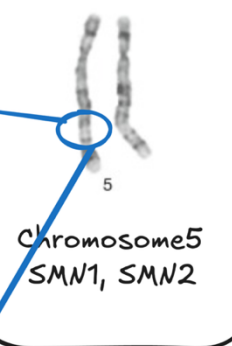
SMN1

Exon6 Intron6 Exon7 Intron7 Exon8

SMN2

Exon6 Intron6 ~~Exon7~~ Intron7 Exon8

mutant



Chromosome5
SMN1, SMN2

ATAATTCCTCCACACCTCCCATATGTCCAGATTCTCTTGATGATGCTGATGCT
TTGGGAAGTATGTTAATTTTCATGGTACATGAGTGGCTATCATACTGGCTAT
TATATG

User가 마음대로 조작 가능!

Next

3. Mature mRNA

SMN1 (정상)

Exon1 Exon2 Exon3 Exon4 Exon5 Exon6 Exon7 Exon8

SMN2 (비정상)

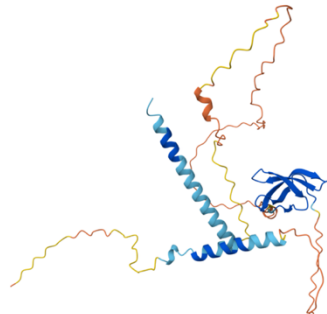
Exon1 Exon2 Exon3 Exon4 Exon5 Exon6 Exon8

No Exon7 !

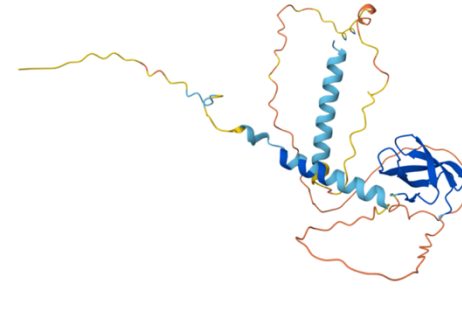
Make Protein

4. Protein structure

SMN1 (정상)



SMN2 (비정상)





SPLICE PLAYGROUND

Visualize Splicing Into Structure

DNA editing, transcript consequences, and protein structure prediction in one
continuous flow

Start

Progress

- Fixing minor errors and adding convenience features.
- The rest of the major framework is complete.

Reference

- Predicting Splicing from Primary Sequence with Deep Learning. *Cell*. 2019 Jan 24;176(3):535-548.e24.
<https://pubmed.ncbi.nlm.nih.gov/30661751/>.
- Jumper, J., Evans, R., Pritzel, A. *et al*. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
<https://doi.org/10.1038/s41586-021-03819-2>.
- <https://towardsdatascience.com/modeling-dna-sequences-with-pytorch-de28b0a05036/>
- Some Lecture Notes from Prof. Daehyun Baek(Seoul National University, School of Biological Sciences)
- <https://www.sciencedirect.com/science/article/pii/S1874939916302759>